Multi-Disease Detection Report

CNN + LLM contextual reasoning • generated 2026-08-24 22:57

Patient context

Age	62
Sex	Male
Symptoms	Blood in urine, persistent flank pain, unexplained weight loss, fatigue
History	62-year-old man, long-term smoker, hypertension, no previous kidney disease

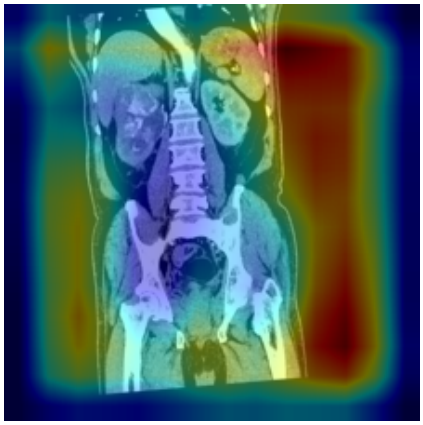
Model prediction

kidney_tumor **Malignant**

Kidney Disease · Renal / Kidney · CT imaging

#	Condition	Confidence
1	kidney_tumor	98.6%
2	kidney_normal	0.3%
3	oral_normal	0.2%
4	Alzheimers_VeryMildDemented	0.2%
5	Alzheimers_MildDemented	0.1%

Grad-CAM (model attention)



LLM contextual interpretation

Model suggests Kidney Disease (kidney_tumor) at 98.6% confidence.

Model finding. The CNN's top class is 'kidney_tumor' (Malignant, Renal / Kidney) on CT imaging, at 98.6%.

Confidence. High model confidence (98.6%) with a clear 98.3 pt margin over the runner-up.

Differential

- **kidney_tumor** — ranked #1 at 98.6%
- **kidney_normal** — ranked #2 at 0.3%
- **oral_normal** — ranked #3 at 0.2%

Grad-CAM correlation. Model attention was a broad, distributed region in the central portion of the image (~36% high-salience area). Verify this region is anatomically plausible for CT imaging.

Biomedical context. [KB] Renal mass/tumor identified on CT. Group: Kidney Disease; organ: Renal / Kidney; modality: CT imaging; category: Malignant (severity: high). Renal masses on CT are evaluated for size,

enhancement, and morphology; solid enhancing lesions raise concern for renal cell carcinoma and warrant further workup. [BioGPT fallback — set ENABLE_BIOGPT=true to run the real model] Renal masses are characterized with dedicated CT/MRI protocols; management ranges from active surveillance to biopsy/resection depending on features. [Med-PaLM 2: PLACEHOLDER — no public API; see medpalm_adapter.py to wire MedLM]

Patient considerations. Provided context (age: 62; sex: Male; symptoms: Blood in urine, persistent flank pain, unexplained weight loss, fatigue; history: 62-year-old man, long-term smoker, hypertension, no previous kidney disease) should be correlated with the finding.

Recommended next steps

- Correlate with the full clinical picture and prior imaging.
- Obtain gold-standard confirmation appropriate to the modality (e.g., biopsy/histology or specialist review).
- Escalate to the relevant specialist if the category is malignant or high-severity.

Uncertainty & limitations

- Interpretation generated WITHOUT the Claude reasoning model (offline fallback).
- Model trained on specific public datasets; performance may not transfer to other scanners/populations.
- A confident softmax score is not calibrated probability of disease.

■ Research decision-support only — not a medical diagnosis. All findings require confirmation by a licensed clinician.